Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Physicist (Ph.D. and M.Sc. in Physics) with ten years of research in computational astrophysics, seeking to teach physics and astronomy. Substitute instructor for a graduate thermodynamics course; mentored three graduate researchers who co-authored papers with me. Bilingual (English/Spanish). Teaching approach built on intuition, critical thinking, and collaborative problem-solving.

Teaching and Mentoring

Thermodynamics (graduate level) — **Jun 2018**
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo — Substitute Instructor

Zachary Davis, Amanda O. Harrison, and Hao Zhang, graduate students — **2018 – 2022**
Department of Physics and Astronomy, Purdue University — Mentoring

Talks and Outreach

El origen del universo y el sistema solar (The Origin of the Universe and the Solar System) — **Feb 2026**
Biology students, Facultad de Ciencias, Universidad Autónoma del Estado de México — Online talk

The Role of Machine Learning in Environmental Science — **Mar 2024**
Seminar on Climate Change Ecology, University of Washington — Seattle, WA

Los más rápidos y los más furiosos (The Fastest and the Most Furious) — **Sep 2020**
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco, Mexico — Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) — **Aug 2020**
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco, Mexico — Online talk

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* — **Jul 2017**

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* — **Sep 2011**

Universidad Autónoma del Estado de México, *B.Sc. in Physics* — **Dec 2008**

Experience

Independent Contractor, *Mercor* — **Jan 2026 – Present**
- Evaluate physics reasoning in AI model outputs, identifying failure modes and providing structured feedback.

Spatial Data Scientist, *TealWaters* — **May 2025 – Jan 2026**
- Mapped wetland probability with a random-forest model (>75% accuracy) and optimized terrain-analysis code.

Independent Researcher & Open-Source Developer, *Self-directed* — **Jan 2024 – May 2025**
- Co-developed Tleco, a Rust/Python toolkit for relativistic-outflow radiation (ApJ 2024).

Software Engineer, *Paychex* — **Apr 2022 – Jan 2024**
- Developed and tested production data pipelines (Java, Python, SQL).

Postdoctoral Research Associate, *Rochester Institute of Technology* — **Feb 2021 – Apr 2022**
- Led an NSF-sponsored upgrade of a black hole binary simulation code; 2 publications on neutron star mergers.

Postdoctoral Research Fellow, *Purdue University* — **Oct 2018 – Nov 2020**
- Co-I on a NASA-funded grant on active-galaxy radiation; mentored three graduate researchers; 3 papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* — **Jan – Sep 2018**
- Built open-source data analysis tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Skills

Physics: Mechanics, electromagnetism, thermodynamics, relativity, statistical physics, astrophysics

Computing: Python, Fortran, C/C++, high-performance/parallel computing

Languages: English (fluent), Spanish (native; talks and teaching in Spanish), Nahuatl (basic)

Selected Publications (9 total, 3 first-author)

Full list available on request.